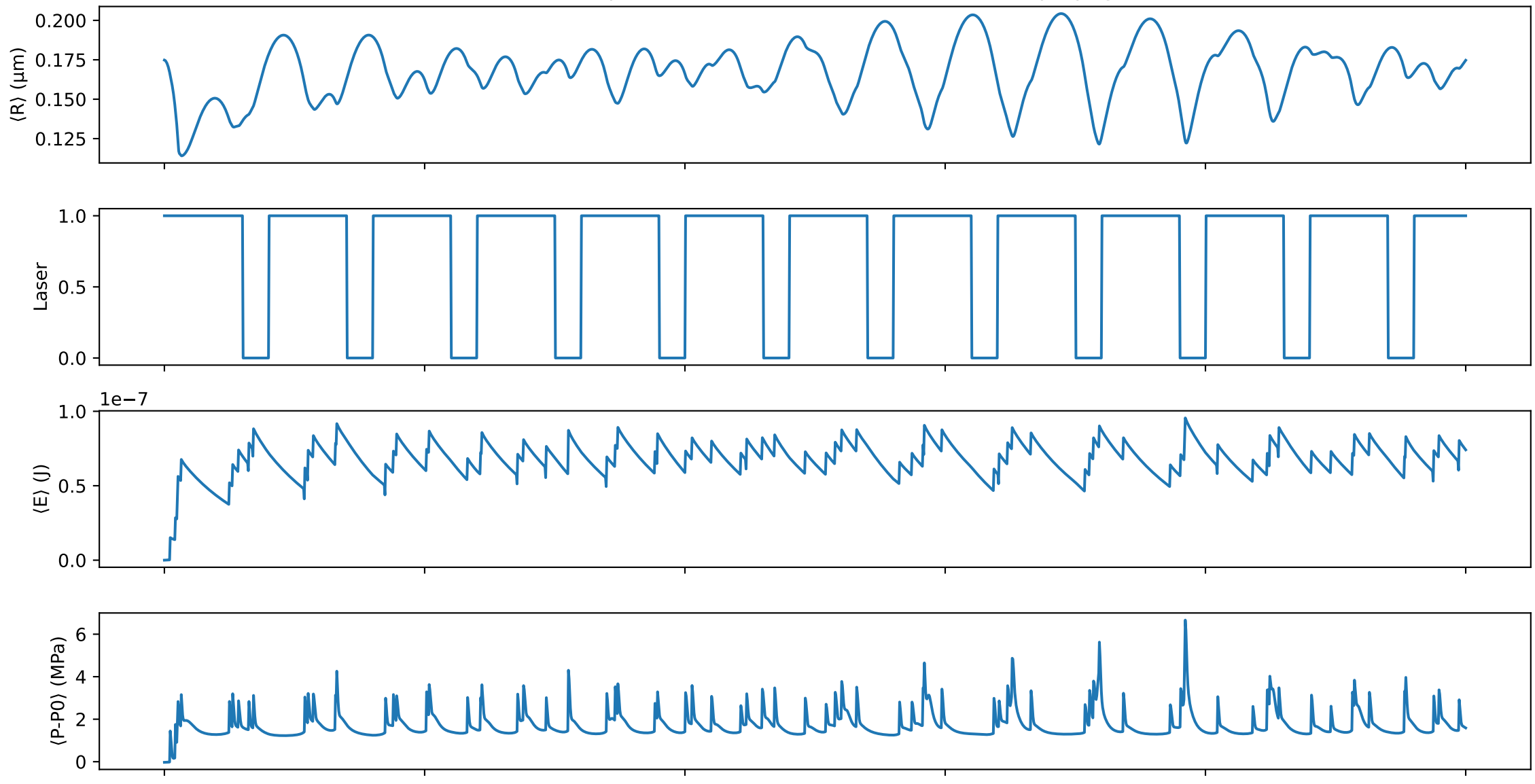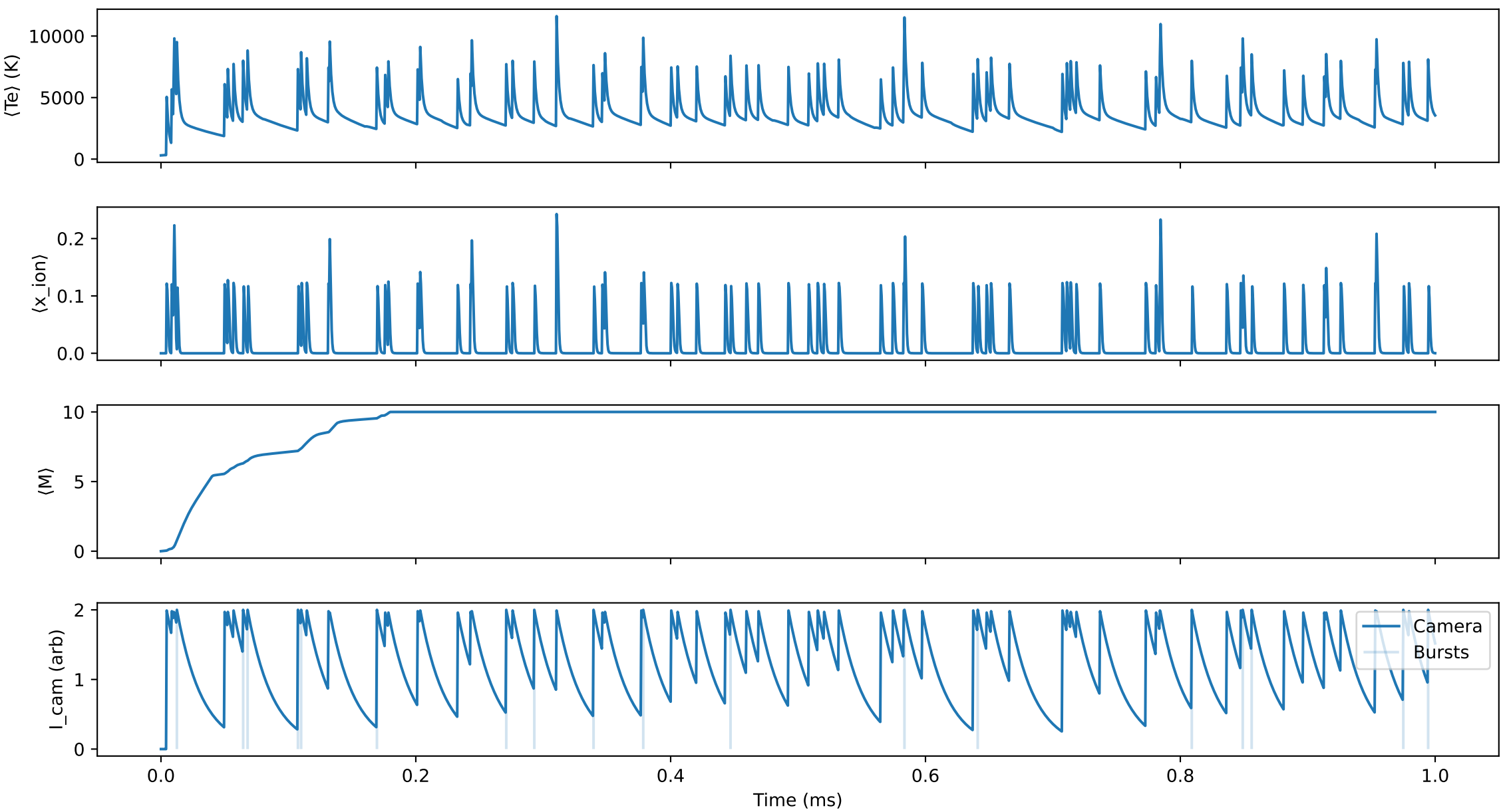Full model: rupture + accretion + Saha EOS + shock propagation